

Ashok Kumar Manohar

Test Architect | AI Quality Engineer | Agentic AI | Playwright | API Automation | CI/CD | RAG & LLM Evaluation

Birmingham, UK | github.com/ashokmanohar-ai | linkedin.com/in/ashok-kumar-manohar

14+	40%	60→95%	<60m
years in Quality Engineering	YoY escaped-defect reduction	FHIR API coverage	regression from 4 hours

PROFESSIONAL SUMMARY

Senior Test Architect and AI Quality Engineering leader with 14+ years of experience designing scalable test automation, CI/CD quality gates, API/UI quality platforms and AI evaluation approaches. Works at the intersection of software quality, Agentic AI, RAG, LLM evaluation, test architecture and developer productivity, with a focus on reliable, auditable and scalable intelligent systems.

CORE EXPERTISE

Test Architecture	Quality Engineering	Agentic AI
Multi-Agent Architecture	AI Agent Evaluation	LLM Evaluation
RAG Evaluation	Prompt Testing	Playwright
Selenium	API Automation	CI/CD
MCP	AI Observability	Quality Governance

PROFESSIONAL EXPERIENCE

Test Architect Mar 2026 - Present
Happiest Minds Technologies

Lead Quality Engineering strategy, enterprise test automation and AI-driven quality initiatives for enterprise and intelligent systems.

- Design automation frameworks using Selenium, Playwright, Python and API testing tools.
- Lead CI/CD quality integration using Jenkins, GitHub Actions and Azure DevOps.
- Develop RAG- and multi-agent-based test design approaches for intelligent test generation, validation and coverage optimisation.
- Drive LLM validation, prompt testing, model reliability and AI system evaluation.
- Define QA standards, automation strategies and engineering best practices; mentor engineers and contribute to continuous improvement.

Quality Lead - Centre of Excellence Dec 2020 - Mar 2026
Acentra Health

Defined enterprise QA, automation and AI-quality standards across healthcare and intelligent systems.

- Achieved a 40% year-on-year reduction in escaped defects through enterprise QA and automation standards.
- Architected FHIR API automation using Rest-Assured, HAPI FHIR, JSON schema validation and synthetic patient-data pipelines, increasing API coverage from 60% to 95%.
- Designed and scaled a Java/Selenium Page Object Model framework with 500+ page objects integrated into CI pipelines, reducing full-suite regression from four hours to under 60 minutes.
- Led DeepEval- and GEval-based LLM evaluation frameworks, reducing LLM regressions by 75%.
- Supported MLOps and LLMOps adoption across AWS, Azure and GCP with CI/CD, governance, monitoring and evaluation gates.

SDET / Automation Tester - evoBrixX Dec 2016 - Dec 2020
Acentra Health

Extended hybrid automation across web, desktop, mainframe and mobile systems.

- Built 500+ reusable Java libraries for complex CRM workflows.
- Reduced manual regression effort by 60% through data-driven automation.
- Prevented 85% of high-severity defect escapes through root-cause analysis and automated smoke checks.
- Trained 15+ testers and maintained release-readiness dashboards.

Delivered quality engineering for banking applications under strict delivery and compliance requirements.

- Managed regression testing for Capital One banking applications.
- Validated IVR workflows covering DTMF and voice recognition.
- Automated ACH/NACHA NOC workflows with 99% audit compliance.
- Improved automation reuse by 50% using QTP/UFT hybrid frameworks.

SELECTED PORTFOLIO

Agentic Quality Engineering Platform

Enterprise multi-agent Quality Engineering platform where specialised AI agents collaborate across requirement analysis, test design, automation generation, execution, review, security and CI/CD.

Focus: Agentic AI · Multi-Agent Systems · Python · Playwright · CI/CD · Quality Governance

Smart Test Designer

AI-powered test design platform combining Retrieval-Augmented Generation with enterprise test engineering workflows to generate contextual, traceable and reusable test scenarios.

Focus: RAG · LLM Evaluation · Python · ChromaDB · Azure OpenAI · REST APIs

Playwright Enterprise Test Framework

Production-oriented Playwright automation framework designed for scalable enterprise UI and API testing with CI/CD integration and evidence-rich reporting.

Focus: Playwright · TypeScript · API Testing · GitHub Actions · CI/CD · Reporting

AI Agent Evaluation Framework

Framework for evaluating AI agents across correctness, task completion, tool usage, reliability, hallucination risk, latency, cost and safety-oriented checks.

Focus: Agent Evaluation · LLM-as-a-Judge · Python · Tool Validation · Reliability · Cost Tracking

RAG & LLM Evaluation Lab

Practical evaluation environment for measuring Retrieval-Augmented Generation quality across retrieval, context relevance, faithfulness and hallucination risk.

Focus: RAG · RAGAS · LLM Evaluation · Embeddings · Reranking · Python

PUBLICATIONS

- Testing MCP-Powered AI Agents: Security, Authorization and Quality Engineering Patterns
- RAG Quality Engineering: A Practical Framework for Evaluating RAG Systems

CERTIFICATIONS

- Machine Learning Operations, MLOps & LLMOps Specialization - Psitron Technologies (Dec 2025)
- Introduction to Cognitive Project Management in AI, CPMIA - PMI (Jul 2025)
- AWS Generative AI Sales - Specialized (Jul 2024)
- AWS Cloud Economics Essentials (Jul 2024)
- AWS Generative AI Essentials (Aug 2023)
- AWS Generative AI Technical - Specialized (Apr 2025)
- Microsoft Certified: Azure AI Fundamentals (Oct 2021)
- AWS GovCloud (US) Essentials (Jul 2024)
- Generative AI Overview for Project Managers - PMI (May 2024)

EDUCATION & AWARDS

Education: B.E., Electronics & Communication Engineering - College of Engineering, Guindy, Chennai (May 2011)

Awards: Lead The Way Award - Acentra Health (2024) · Best Rookie of the Year - Wipro Technologies

TECHNOLOGY

OpenAI · Azure OpenAI · Claude · RAG · DeepEval · GEval · RAGAS · LLM-as-a-Judge · Playwright · Selenium · Java · Python · TypeScript · JavaScript · Cucumber · Rest-Assured · REST · OpenAPI · GraphQL · GitHub Actions · Jenkins · Azure DevOps · Docker · Kubernetes · AWS · Azure · GCP · OpenTelemetry